PSNA College of Engineering and Technology, Dindigul – 624 622.
(An Autonomous Institution Affiliated to Anna University, Chennai)
FIRST SERIAL TEST :: 2026-27:: ODD Semester -Answer Key

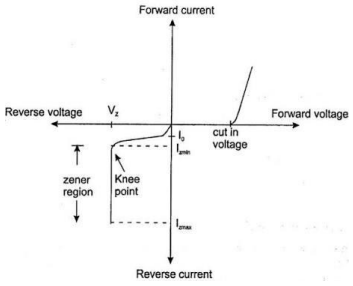
B.E Biomedical engineering — Max. Marks 60
Semester III — 2022 — Date:25.08.2026
BM2312 | Electronic Devices & Circuits — Time :2 Hours

No **Part–A: Answer All Questions.** **10 x 2 = 20 Marks**

1 **Explain the difference between intrinsic and extrinsic semiconductors**.

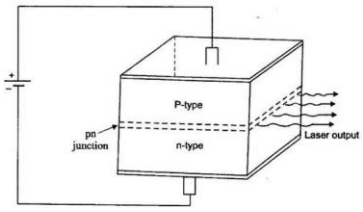
Intrinsic Semiconductor	Extrinsic Semiconductor
Pure semiconductor without impurities.	Semiconductor doped with impurities.
Number of electrons and holes are equal.	Majority and minority charge carriers exist.
Example: Pure silicon and germanium.	Types: P-type and N-type semiconductor.

2 **Draw the V-I characteristics of Zener Diode and explain its operation.**



A Zener diode operates like a normal diode in forward bias. In reverse bias, only a small current flows until the reverse voltage reaches the Zener breakdown voltage VZV_Z. Beyond this point, the current increases rapidly while the voltage across the diode remains nearly constant.

3 **Explain the principle behind the LASER diode with a neat sketch.**



A LASER diode works on the principle of stimulated emission. When forward biased, electrons and holes recombine in the active region and produce photons. These photons stimulate further emissions, producing coherent, monochromatic and highly directional light. The optical cavity amplifies the light, and a laser beam emerges from one partially reflecting surface.

4 **When a reverse bias is applied to a germanium PN junction diode, the reverse saturation current is 0.3×10^{-6} A at room temperature. Determine the current flowing in the diode when 0.15 V forward bias is applied at room temperature.**

I =120.73μA

5 **What is a bipolar junction transistor? How are its terminals named?**

A Bipolar Junction Transistor (BJT) is a three-layer, three-terminal semiconductor device in which both electrons and holes participate in conduction. It is mainly used for amplification and switching.
Its three terminals are:

- Emitter (E) – supplies charge carriers.
- Base (B) – controls the flow of charge carriers.
- Collector (C) – collects the charge carriers.

6 **Explain the construction of N channel JFET.**

An N-channel JFET consists of a narrow N-type semiconductor channel with two ohmic contacts called Source (S) and Drain (D). Two heavily doped P-type regions are formed on either side of the N-channel and connected together to form the Gate (G) terminal.

The current flows through the N-channel from source to drain, and the gate voltage controls the channel width.

7 **Describe the operation of a Shockley Diode.**

A Shockley diode is a four-layer PNPN, two-terminal semiconductor device. It remains in the OFF state until the applied forward voltage reaches the breakover voltage. Beyond this voltage, it switches suddenly to the ON state and conducts heavily.

It continues conducting until the current falls below the holding current.

8 **The transistor has I_E =10mA and α=0.98. Determine the values of I_C and I_B**

IC = 9.8mA IB=0.2mA

9 **What is meant by Q-point?**

The Q-point (Quiescent Point) is the operating point of a transistor when no input signal is applied. It represents the steady values of collector current (IC) and collector-emitter voltage (VCE)determined by the DC load line.

10 **Determine the h-parameters from the characteristics of CB configuration**

$$h_{ib} = \left(\frac{\Delta V_{EB}}{\Delta I_E}\right)_{V_{CB}=\text{constant}} \qquad h_{fb} = \left(\frac{\Delta I_C}{\Delta I_E}\right)_{V_{CB}=\text{constant}}$$

$$h_{rb} = \left(\frac{\Delta V_{EB}}{\Delta V_{CB}}\right)_{I_E=\text{constant}} \qquad h_{ob} = \left(\frac{\Delta I_C}{\Delta V_{CB}}\right)_{I_E=\text{constant}}$$

Part–B: Answer All Questions. (2*16)+ (1*8) = 40 Marks

11 A (i) **Apply the concept of PN-junction biasing to explain the operation of a diode under forward-bias and reverse-bias conditions. Illustrate the changes in depletion-region width, barrier potential and current flow with suitable circuit diagrams and V–I characteristics**.

1. **Introduction to PN-Junction Biasing**

A PN junction is formed by joining a P-type semiconductor with an N-type semiconductor. At the junction, electrons from the N-region and holes from the P-region recombine, forming a depletion region containing immobile ions. This depletion region produces an internal electric field and a barrier potential that opposes the movement of majority charge carriers. By connecting an external DC supply to the PN junction, its operation can be controlled. This process of applying an external voltage to a PN junction is called biasing. Depending on the polarity of the applied voltage, the diode operates in either forward bias or reverse bias.

The PN-junction diode and its rectifying behavior form the basis of the rectifier circuits discussed in the supplied material.

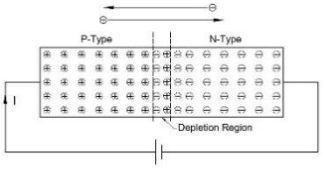
2**. PN Junction at Zero Bias**

When no external voltage is applied to the PN junction, the diffusion of majority carriers across the junction causes electrons and holes to recombine near the junction. This produces a depletion region. The positive donor ions on the N-side and negative acceptor ions on the P-side establish an electric field across this region. This electric field produces the barrier potential, which prevents further majority-carrier diffusion under equilibrium conditions.

The depletion region therefore acts as a barrier between the P and N regions. When an external voltage is applied, the width of this region and the barrier potential change depending on the polarity of the applied voltage.

3. **Forward-Bias Operation**

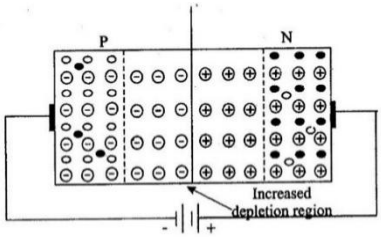
In forward bias, the P-side of the PN junction is connected to the positive terminal of the supply and the N-side is connected to the negative terminal.



The applied voltage opposes the internal electric field of the depletion region. Therefore, the width of the depletion region decreases and the barrier potential is reduced. As the applied forward voltage is increased, majority carriers acquire sufficient energy to cross the junction. Holes move from the P-side towards the N-side, while electrons move from the N-side towards the P-side. Consequently, the diode current increases rapidly after the cut-in voltage is reached. Thus, in forward bias, the diode offers a low resistance to current flow. This unidirectional conduction property is the basis for the operation of rectifier circuits.

4. Reverse-Bias Operation

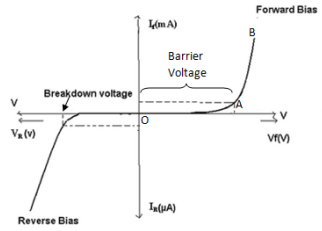
In reverse bias, the P-side is connected to the negative terminal and the N-side is connected to the positive terminal of the supply.



The applied voltage acts in the same direction as the internal electric field of the junction. Therefore, the depletion-region width increases and the barrier potential increases. The majority charge carriers are pulled away from the junction, so they cannot cross the depletion region. Hence, the diode offers a high resistance to current. Only a very small reverse current due to minority carriers flows through the diode. If the reverse voltage is increased beyond the breakdown voltage, the reverse current increases sharply.

5. V–I Characteristics of PN-Junction Diode

The V–I characteristic represents the relationship between the voltage applied across the diode and the resulting current. In the forward-bias region, the current remains very small initially because the barrier potential opposes carrier movement. When the applied voltage reaches the cut-in or knee voltage, the current increases rapidly with a small increase in voltage. In the reverse-bias region, only a small reverse saturation current flows. When the reverse voltage reaches the breakdown voltage, the reverse current increases rapidly. Therefore, the V–I characteristic clearly demonstrates the rectifying nature of the PN-junction diode.



The supplied material describes a rectifier as a device that offers low resistance in one direction and high resistance in the opposite direction, producing a unidirectional output from a sinusoidal input.

OR

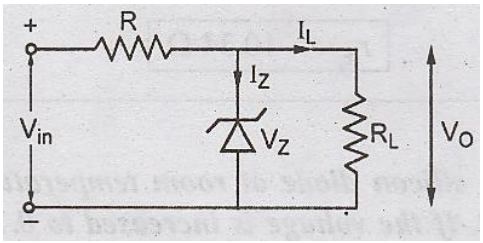
B **(i) Demonstrate the behavior of the voltage regulator when the input voltage and load conditions vary, and explain how the Zener diode maintains a nearly constant output voltage.**

1. **Zener Diode as a Voltage Regulator — 2 Marks**

A Zener diode is operated in its reverse-bias breakdown region to regulate voltage. In this region, the voltage across the Zener diode remains nearly constant even when its current changes considerably, provided that the Zener current remains between the minimum and maximum permissible values.

The Zener diode is connected in parallel with the load, while a series resistor is used to limit the current.

2. Basic Zener Voltage Regulator Circuit — 3 Marks



Here,

- Vin = Input voltage
- R = Series current-limiting resistor
- VZ = Zener voltage
- RL = Load resistance
- Vo = Regulated output voltage

Since the Zener diode and load are connected in parallel,

$V_o \approx V_Z$

The series resistor absorbs the excess input voltage and limits the current through the Zener diode. The supplied material specifically notes that the Zener current is usually limited by a series resistor.

3. Normal Regulating Operation — 2 Marks

When the input voltage is sufficient to place the Zener diode in its breakdown region, the Zener maintains an almost constant voltage across itself. Since the load is connected in parallel with the Zener, $V_o \approx V_Z$

Therefore, even though the Zener current may vary, the output voltage remains nearly constant. The regulator continues to operate as long as the Zener current does not fall below IZ(min) or exceed its safe maximum value.

4. Variation in Input Voltage — 3 Marks

Consider an increase in input voltage.

When Vin increases, the current through the series resistor RS increases. The additional current is mainly diverted through the Zener diode. Thus, the Zener current increases, but the voltage across the Zener changes only slightly because it is operating in the breakdown region.

Therefore, $V_{in}\uparrow \Rightarrow I_Z\uparrow \Rightarrow V_o \approx V_Z$

Similarly, when the input voltage decreases, the current through the series resistor decreases. The Zener current decreases, but the output voltage remains approximately equal to VZ, provided the Zener current remains above IZ(min).

Thus, the Zener diode absorbs the variations in input voltage and protects the load from large changes in supply voltage.

5. Variation in Load Conditions — 3 Marks

When the load resistance changes, the load current also changes. If the load current decreases, more of the current supplied through RS flows through the Zener diode. Hence, IZ increases while Vo remains nearly constant.

If the load current increases, more current is supplied to the load and the Zener current decreases. The Zener continues to regulate the voltage as long as its current does not fall below IZ(min)I_Z(min).

Therefore: $I_L\downarrow \Rightarrow I_Z\uparrow$ and $I_L\uparrow \Rightarrow I_Z\downarrow$ while $V_o \approx V_Z$

The uploaded material states that the regulator is intended to provide a constant output voltage despite variations in supply voltage or load current, and that regulation continues until the Zener current falls below the minimum value required for breakdown operation.

6. **Regulation Range— 3 Marks**

The Zener diode can maintain regulation only within its specified operating range. The Zener current must remain between the minimum breakdown current IZ(min) and the maximum permissible Zener current IZ(max) If the current falls below IZ(min) the diode leaves the breakdown region and voltage regulation is lost. If the current becomes excessively high, the Zener diode may be damaged.

Thus, the Zener diode acts as a shunt voltage regulator by maintaining an approximately constant voltage across the load despite changes in input voltage and load current.

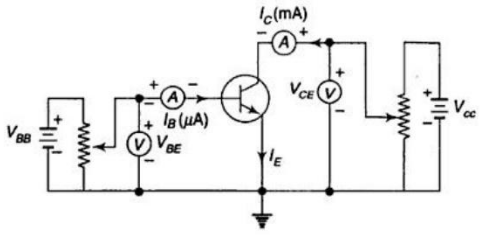
$V_o \approx V_Z$

12 A **(i) Discuss the input and output characteristics of a Common Emitter (CE) transistor configuration with neat characteristic curves. Define and explain the transistor parameters obtained from these characteristics.**

1. CE Configuration and Basic Circuit — 2 Marks

In a Common Emitter (CE) configuration, the emitter terminal is common to both the input and output circuits. The input is applied between base and emitter, while the output is taken between collector and emitter. The emitter is generally connected to the common reference point. The input current is IBI_B, output current is IC, input voltage is VBE, and output voltage is VCE.

The CE configuration is widely used for voltage and current amplification.



2. Input Characteristics of CE Configuration — 3 Marks

The input characteristic of a CE transistor is the relationship between base current IBand base-emitter voltage VBE at constant collector-emitter voltage VCE

IB=f(VBE)at constant VCE

The characteristic is similar to the forward characteristic of a PN-junction diode because the base-emitter junction is forward biased. Initially, the base current is very small. As VBE reaches the cut-in voltage, the base current increases rapidly with a further increase in VBE. For a silicon transistor, the cut-in voltage is approximately 0.7 V.

Input Characteristic

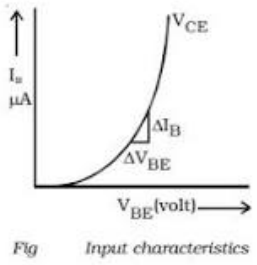


Fig Input characteristics

The input characteristic is useful for determining the input resistance of the transistor.

3. Output Characteristics of CE Configuration — 3 Marks

The output characteristic represents the relationship between collector current IC and collector-emitter voltage VCE for different constant values of base current IB.

IC=f(VCE)at constant IB

The output characteristics contain three important operating regions: cut-off, active and saturation. In the cut-off region, the base current is approximately zero and the transistor is OFF. In the active region, the collector current is mainly controlled by the base current and remains almost constant with changes in VCE. This is the region normally used for amplification. In the saturation region, both junctions are forward biased and the transistor operates in the ON state.

Output Characteristic

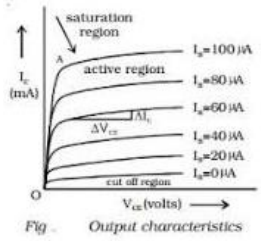


Fig Output characteristics

Thus, the output characteristic helps identify the operating region and determine the transistor's output parameters.

4. Input Resistance (hie) — 2 Marks

The input resistance is the ratio of the small change in input voltage to the corresponding change in input current, with the output voltage kept constant.

For CE configuration,hie=(ΔVBE/ΔIB) VCE=constant

It is obtained from the slope of the input characteristic.

Unit: Ohm (Ω)

A steeper input characteristic indicates lower input resistance, whereas a flatter characteristic indicates higher input resistance.

5. Forward Current Gain (hfe) — 2 Marks

The forward current gain represents the ratio of the change in collector current to the corresponding change in base current, while VCE is kept constant.

hfe=(ΔICΔIB) VCE=constant

It is obtained from the output characteristic.

It is dimensionless.

6. Reverse Voltage Ratio and Output Admittance — 2 Marks

The reverse voltage ratio hreh_{re} represents the change in input voltage due to a change in output voltage when the input current is kept constant.

hre=(ΔVBEΔVCE)IB=constant

It is dimensionless and generally has a small value.

The output admittance hoe represents the change in collector current with respect to collector-emitter voltage when the base current is kept constant.

hoe=(ΔICΔVCE)IB=constant

Its unit is siemens (S) or Ω^{-1}

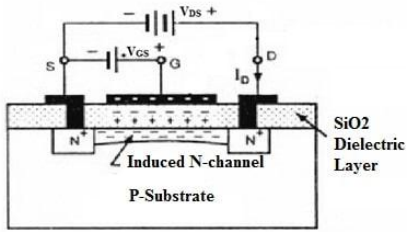
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B **(i) Explain the construction and operation of an N-channel enhancement MOSFET. Describe the formation of the inversion channel and explain its drain characteristics in cut-off, linear and saturation regions with suitable graphs**

1. Construction of N-Channel Enhancement MOSFET — 3 Marks

An N-channel enhancement MOSFET (E-MOSFET) is a voltage-controlled semiconductor device in which the conducting channel is not physically present at zero gate voltage. It consists of a lightly doped P-type substrate into which two heavily doped N+ regions are formed. These two N+ regions act as the Source (S) and Drain (D). A thin insulating layer of silicon dioxide (SiO_2) is placed over the region between source and drain, and a metallic Gate (G) electrode is formed above the oxide layer. The gate is therefore electrically insulated from the semiconductor.

Construction diagram

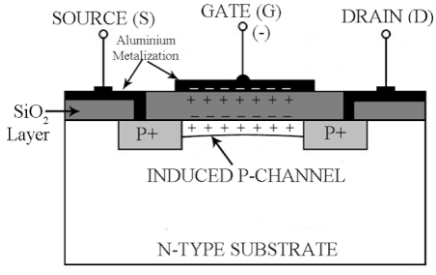


The four terminals are Gate, Drain, Source and Body/Substrate. In many practical circuits, the body is connected to the source.

2. Formation of the Inversion Channel — 3 Marks

When VGS=0, there is no conducting N-channel between the source and drain. Therefore, the drain current is almost zero, apart from a very small leakage current. When a positive voltage is applied between gate and source, an electric field is produced through the oxide layer. This positive gate voltage repels holes away from the surface of the P-type substrate and attracts electrons towards the region below the gate. As VGS is increased, the concentration of electrons near the surface increases.

When VGS reaches the threshold voltage VTH, sufficient electrons accumulate near the surface to form an N-type inversion layer connecting the source and drain. This conducting layer is called the inversion channel or induced N-channel.



Thus, unlike a depletion MOSFET or JFET, the N-channel enhancement MOSFET creates its channel only when the applied gate voltage exceeds the threshold voltage.

3. Operation of N-Channel Enhancement MOSFET — 3 Marks

For operation, a positive VDS is applied with the drain positive with respect to the source. When

VGS<VTH

the inversion channel is not formed and the MOSFET remains OFF. Therefore,ID≈0 .When VGS>VTH an N-channel is induced between the source and drain. Applying VDSV_{DS} causes electrons to move from the source towards the drain through the induced channel, producing a drain current ID. Increasing VGS increases the strength of the inversion channel and hence increases the drain current.

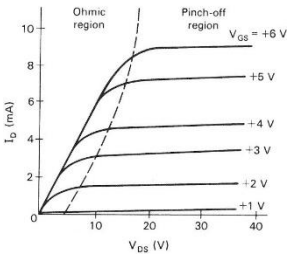
Therefore, the MOSFET is controlled by gate voltage rather than gate current, because the oxide layer electrically isolates the gate from the channel.

4. Drain Characteristics — 2 Marks

The drain characteristic of an N-channel enhancement MOSFET represents the relationship between drain current IDI_D and drain-source voltage VDS} for different constant values of VGS.

ID=f(VDS)at constant VGS .A family of curves is obtained by varying VGS.

Drain Characteristics



The characteristic consists mainly of cut-off, linear (ohmic) and saturation regions.

5. Operating Regions — 3 Marks

(a) Cut-off Region

When VGS<VTH the inversion channel is not formed. Consequently, there is practically no conduction between source and drain and ID≈0 The MOSFET behaves approximately like an open switch.

(b) Linear / Ohmic Region

When VGS>VTH and VDS is relatively small, an inversion channel exists throughout the region between source and drain. The drain current increases approximately with VDS. The MOSFET behaves like a voltage-controlled resistance in this region.

As VDS increases, the channel becomes narrower near the drain because the effective gate-to-channel voltage is reduced near the drain.

(c) Saturation Region

As VDS is increased further, the channel becomes progressively narrower near the drain. At the boundary of saturation, VDS=VGS−VTH Beyond this condition, the channel is pinched off near the drain end. Further increases in VDS produce only a relatively small increase in IDI_D. The MOSFET therefore operates in the saturation region, which is commonly used for amplification.

For an ideal long-channel MOSFET,

ID=12k(VGS−VTH)2

in saturation, where kk is the device transconductance parameter.

13 A **(i) Explain the concept of small-signal analysis of a BJT amplifier. Draw the small-signal equivalent circuit of a BJT using the hybrid-π model and explain the significance of its parameters.**

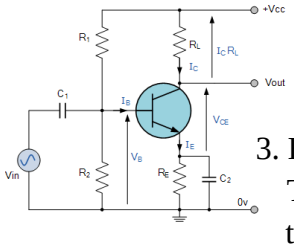
1. Concept of Small-Signal Analysis — 3 Marks

A BJT amplifier is normally biased at a suitable Q-point (quiescent point) so that the transistor operates in the active region. When a small AC signal is applied around this operating point, the variations in transistor voltage and current are small. The transistor can therefore be represented by a linear equivalent circuit around the Q-point. This method of replacing the nonlinear BJT with an approximate linear model for small variations is called small-signal analysis.

The purpose of small-signal analysis is to determine important amplifier quantities such as voltage gain, current gain, input resistance and output resistance without directly analysing the nonlinear transistor characteristics.

2. BJT Amplifier and Operating Point — 2 Marks

Before performing small-signal analysis, the BJT is biased so that its Q-point lies in the active region. The total transistor voltage and current consist of DC and small AC components.

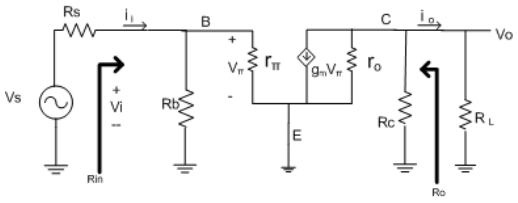


Tthe small-signal model represents the behaviour of the transistor only for small variations around the Q-point.

3. Hybrid-π Small-Signal Equivalent Circuit — 4 Marks

The hybrid-π model is a widely used small-signal equivalent model of a BJT. In this model, the base-emitter junction is represented by an internal resistance rπr_\pi, while the controlled collector current is represented by a dependent current source.

For a CE amplifier, the simplified hybrid-π model is:



Here,

vπ=vbe -is the small-signal voltage between base and emitter. The dependent current source represents the variation of collector current produced by the small change in base-emitter voltage.

4. Important Hybrid-π Parameters — 5 Marks

(a) Base-Emitter Resistance rπ

The parameter rπ represents the small-signal input resistance between base and emitter.

rπ=vbe/ib

It determines how much input current flows for a given small-signal input voltage.

A higher rπi generally results in a higher input resistance of the transistor stage.

(b) Transconductance gm

The transconductance represents the change in collector current produced by a change in base-emitter voltage. gm=ic/vbe

For a BJT, gm=IC/VT

where IC is the quiescent collector current and VT is the thermal voltage.

At room temperature, VT≈25 mV

Thus, gm indicates the ability of the transistor to convert an input voltage variation into an output current variation.

(c) Output Resistance ro

The output resistance ro represents the small-signal resistance looking into the collector due to the Early effect.

It is approximately given by ro≈VA/IC

where VA is the Early voltage.

A large ro indicates that the collector current is less affected by changes in VCEV_{CE}.

6. Advantages and Applications of Small-Signal Analysis — 2 Marks

Small-signal analysis provides a simple method for analysing BJT amplifiers around their operating point. It avoids repeated use of the nonlinear transistor characteristic and allows the amplifier's gain, input impedance and output impedance to be determined using standard circuit-analysis techniques.

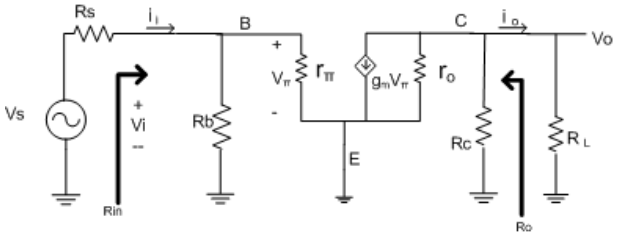
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B (i) **Explain the effect of source resistance, load resistance and transistor parameters on the voltage gain of a common-emitter BJT amplifier. Derive the expression for overall voltage gain.**

1. CE Amplifier and Small-Signal Model — 2 Marks

A common-emitter (CE) amplifier is a BJT amplifier in which the emitter terminal is common to both the input and output circuits. The input signal is applied between the base and emitter, and the output is obtained between the collector and emitter.

For small-signal analysis, the BJT can be represented by the hybrid-π model.



The important small-signal parameters are rπ, gm, and ro.

2. Effect of Source Resistance RSR_S — 3 Marks

The signal source has a source resistance RS, while the amplifier has an input resistance Rin.

The input voltage actually applied to the amplifier is obtained by voltage division:

$Vi=Vs[Rin/RS+Rin]$

Therefore,

$Vi/Vs=Rin/RS+Rin$

If RS increases, the voltage drop across RS increases and less voltage reaches the amplifier.

Hence, $RS\uparrow \Rightarrow Vi\downarrow \Rightarrow Av(overall)\downarrow$

Therefore, source resistance reduces the overall voltage gain due to input voltage-divider action.

3. Effect of Load Resistance RLR_L — 3 Marks

The load resistance RL is connected to the output of the CE amplifier. It appears in parallel with the collector resistance.

The effective collector resistance is $RC'=RC\|RL$

If transistor output resistance ro is considered,

$RC'=RC\|RL\|ro$

The CE voltage gain is approximately $Av=-gmRC'$

Therefore, when RL decreases, $RC'\downarrow$ and consequently, $|Av|\downarrow$ Thus, $RL\downarrow \Rightarrow RC'\downarrow \Rightarrow |Av|\downarrow$

A low load resistance loads the amplifier heavily and reduces its voltage gain.

4. Effect of Transistor Parameters — 3 Marks

The voltage gain is strongly influenced by the transistor's small-signal parameters.

(a) Transconductance gm

The transconductance is $gm=IC/VT$

where IC is the quiescent collector current and VT is the thermal voltage.

Since

$|Av|\approx gmRC'|$ an increase in gm increases the magnitude of voltage gain.

(b) Input resistance rπ

The hybrid-π input resistance is $r\pi=\beta/gm$

It determines the input resistance of the transistor and therefore affects the voltage division between RS and the amplifier input.

A higher rπ generally allows a greater portion of the source voltage to appear at the transistor input.

(c) Output resistance ro

Due to the Early effect, the transistor has a finite output resistance: $ro\approx VA/IC$

Therefore,

$RC' = RC \| RL \| ro$

A finite ro reduces the effective collector resistance and hence reduces the voltage gain compared with the ideal case.

5. Derivation of CE Stage Voltage Gain — 3 Marks

From the hybrid-π model,

$ic = gmv\pi$

The effective collector resistance is $RC' = RC \| RL \| ro$

The output voltage is $vo = -icRC'$

Substituting $ic = gmv\pi i$,

$vo = -gmv\pi RC'$

For the CE amplifier, $vi \approx v\pi$

Therefore, the voltage gain of the amplifier stage is $Av,stage = \frac{vo}{vi}$

Hence,

$Av,stage = -gm(RC \| RL \| ro)$,If ro is neglected, $Av,stage \approx -gm(RC \| RL)$

The negative sign indicates that the CE amplifier produces a 180° phase reversal between input and output.

6. Overall Voltage Gain — 2 Marks

The overall voltage gain is

$Av = \frac{Vo}{Vs}$

It can be written as

$\frac{Vo}{Vs} = \frac{Vo}{Vi} \times \frac{Vi}{Vs}$

We already have

$\frac{Vo}{Vi} = -gm(RC \| RL \| ro)$

and from the input voltage-divider,

$\frac{Vi}{Vs} = \frac{Rin}{RS + Rin}$

Therefore,

$Av = -gm(RC \| RL \| ro)\frac{Rin}{RS + Rin}$

For a simplified CE amplifier where $Rin \approx r\pi$,

$Av = -gm(RC \| RL \| ro)\frac{r\pi}{RS + r\pi}$

If ro is neglected,

$Av \approx -gm(RC \| RL)\frac{Rin}{RS + Rin}$